

Deutsche Telekom (DTE) – LONG

TARGET PRICE: €50.11 (38.49% UPSIDE)



Cambridge Student Investment Fund

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Commodity Product, Oligopolistic Industry, CapEx as barrier to entry, Consistent Growth Trajectory

● Mode of Competition

- Commoditized product – in theory – mostly price
- However, prices are generally similar, therefore:
 1. Brand Rating/Trust
 2. Network Speed (mbps) – 4g/5g
 3. Network Coverage

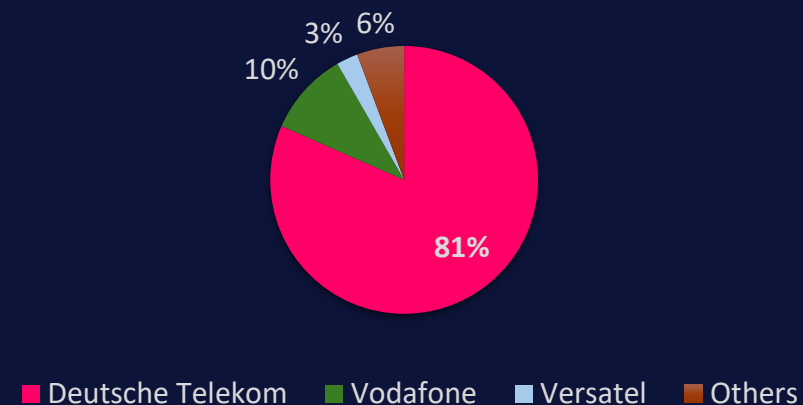
● Industry Outlook

- Projected European Market Growth $\approx$ 1.7% partly because of high levels of regulation
- Projected US market growth $\approx$ 3.7%
- Impacts of AI are positive – reduces costs and increases demand.
- Growth is non-cyclical i.e. “more for more”

● Competitive Landscape

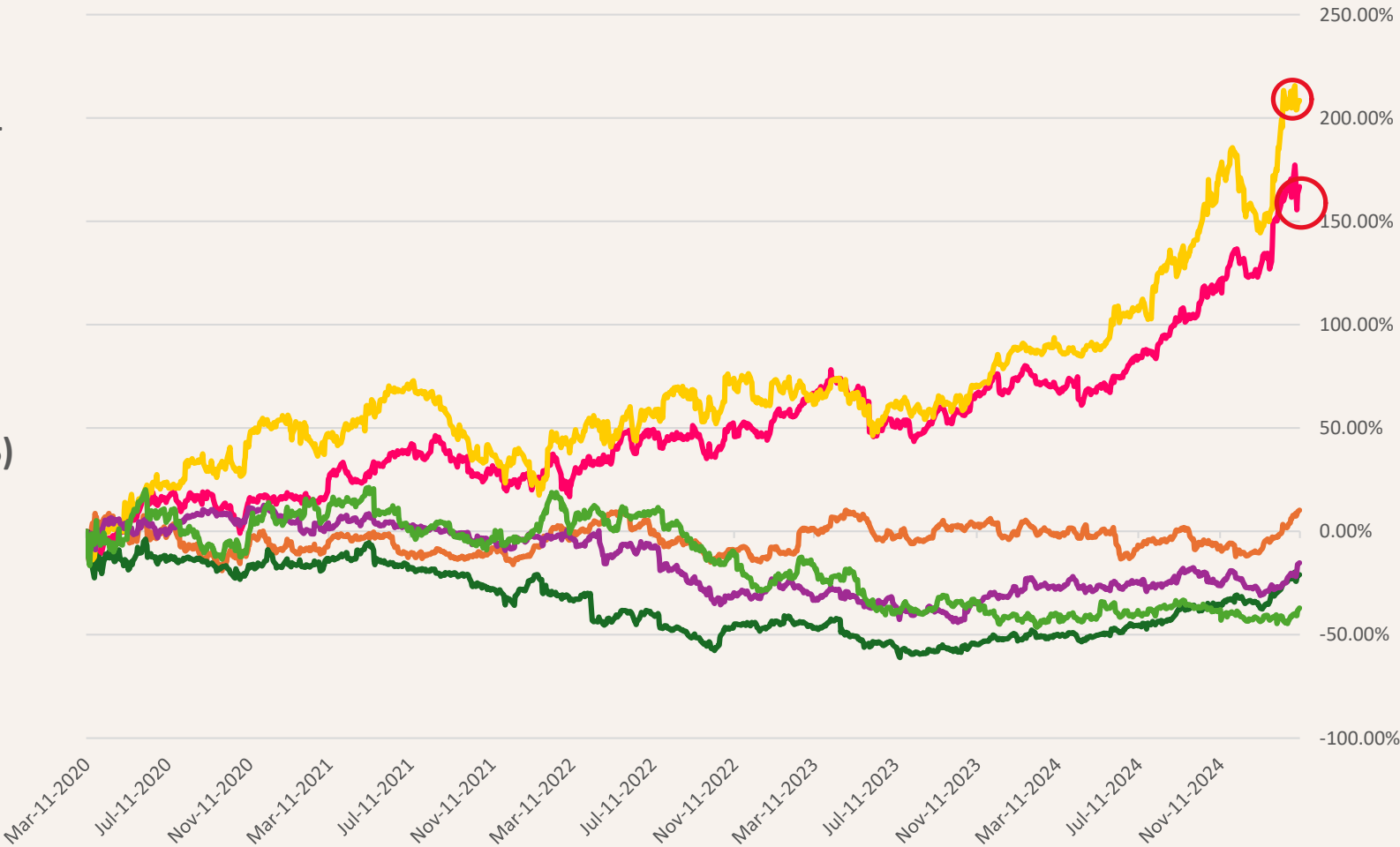
- Main competitors in Germany: Vodafone, Telefónica, RTL
- Main competitors in USA: Verizon, AT&T, Comcast
- German market – full dominance, high margin.
- US market - the largest in the world
 - US market size $\approx$ \$560 bn vs China $\approx$ \$430 bn

German Market Share



DTE and TMUS are the only value-creating large-cap telecom companies in the past 5 years

- Deutsche Telekom AG (XTRA:DTE) - Share Pricing
- Orange S.A. (ENXTPA:ORA) - Share Pricing
- AT&T Inc. (NYSE:T) - Share Pricing
- T-Mobile US, Inc. (NasdaqGS:TMUS) - Share Pricing
- Verizon Communications Inc. (NYSE:VZ) - Share Pricing
- Vodafone Group Public Limited Company (LSE:VOD) - Share Pricing



COMPANY OVERVIEW

DTE unique business model explains its stock performance

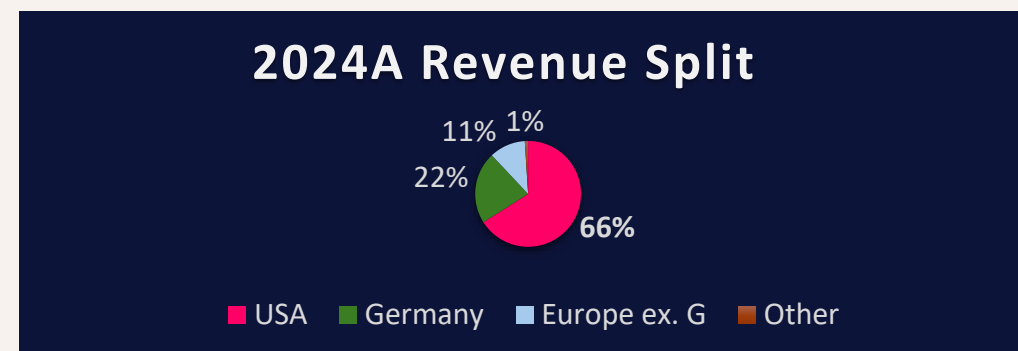
Cap Table		Summary of Key States & KPIs			
		Years	2022A	2023A	2024A
Share Price	€33.81				
(x) FDSO	4,891.4				
Market Cap	€165,379	Revenue	117,074	114,733	118,432
		%y/y	6.0%	(2.0%)	3.2%
		Gross Profit	45,951	49,769	52,952
(+) Debt	152,439	%margin	39.2%	43.4%	44.7%
		%y/y	8%	6%	3%
(-) Cash	8,478	Adj. EBIT	14,546	21,994	24,399
EV	€344,684	%margin	12.4%	19.2%	20.6%
		%y/y	51%	11%	12%
Total Shares Public	3,375.1	Unl. FCF	13,617	19,614	19,991
%float	67.7	%y/y	223%	44%	3%

● Business Model (GER Cash Cow, US Star)

- Previously a state-owned monopoly (not inherited the debts!)
- The company offers a comprehensive portfolio of **fixed-network** and **mobile** communications, **broadband** solutions, and IT services.
- **In Germany – biggest incumbent**, strong profits, the company strategy is not to grow in Germany but to extract by offering FTTH.
- The **core strategy** is to grow in the US, **keep raising its stake in TMUS** which pursues an aggressive inorganic growth strategy.

● Geographic Breakdown

CEO: *“the **US** and our **European** entity gives us a super advantage to scale our businesses”*



● DT has an unparalleled network quality

- **Germany:** Integrated provider of fixed-network and mobile services. **Winner of all major quality awards.**
- **USA:** Ookla – *“T-mobile wins awards for the **fastest**, most **consistent**, and **most available** 5G network, alongside **highest-ranking consumer sentiment**”*
- 5G network covers 98% of German population and 78% of Europe. **No.1 5G (coverage and speed) in all its markets.**

Thesis #1 – Superb Bottom-Line Performance

Management's calculated bets have paid off - DTE overdelivered on all financial and operating targets.

● Hit and/or exceeded all 2020 AGM targets

- Financial: TMUS stake >50%, divestments, deleveraging to <2.75x, 50% EPS dividend. Operating: FTTH rollout in Germany, top transatlantic market share. **All targets exceeded.**
- Recall this was period of higher energy costs. DTE started expensive 'insourcing' for resilience, and COVID deployment disruption. **Yet, all segments outperformed EBITDA targets.**
- **Most telcos failed to hit financial targets and created no value.**

● The Promise: Superb Bottom-Line Performance

- DT has done the hard yards of investment, now expect expanding margins as capital intensity eases, especially in U.S.
- **FCF, 30% CAGR from 2020 to 2024**, now exceeding €18 billion annually. **EPS grew at 10% CAGR in recent years.**
- There are only 10 European large caps that offer DD EPS growth. They trade at 23x P/E; DTE trades at 15x P/E.

● T-Mobile is an expert at inorganic growth

- Sprint merger huge success to penetrate FWA market.
- Sprint was the 4th largest telecom in the US; that acquisition has fueled T-Mobile's EBITDA performance for the past 5 years.
- **This was not a one off - TMUS has completed 12 fibreco M&A deals since 2023**— these are complex and resource intensive.
- They **set the grounds for a Sprint-style acquisition of a major player, this time in the fibre market.** Market misunderstands; has not priced in this 'larger plan'.
- Pending acquisitions (all US fibre): USM, Lumos and Metronet. Zero upside is priced in the company's internal estimates.
- **JPM estimates that by 2030E they alone could raise TMUS revenues 9%, EBITDA 7%, FCF 10%.**

Thesis #2 – Financials & Portfolio Mix = Defensive Play

Strong financials, revenue diversification and prior investment have given DTE a competitive edge

● Significant exposure to the US market

- 51.54% majority stake in T-Mobile US, the #1, fastest growing US telecommunications company
- **T-Mobile trades at 27x P/E but DT trades at 15x P/E. Yet, 75% of DT earnings are earnings from T-Mobile.**
- 2020 T-Mobile and Sprint merger increased Deutsche Telekom's revenue from the US segment.

● Stronger financials compared to competitors

- High EBITDA margin (34.3%) (compared to Vodafone – 24.4%, Telefónica - 23.6%, Orange - 32.2%) signals strong operations.
- Deutsche Telekom has grown revenues at a **6% CAGR since 2015** and a **3.5% CAGR since 2020, outperforming all its peers.**
- Cash pile is €1.45 billion, **cost of debt (~5%) is amazing** too.

● Dominant position in domestic market

- Largest telecommunications service provider in Germany. Strong domestic position: 40% mobile subscriber share and 31% broadband share
- Strong financials enable investment into FTTH lines which will increase broadband revenue and customer base.
- **Notably other German operators did not invest in the face of uncertainty and financial issues, so DTE gained market share.**

● Uniquely Diversified Revenues

- Geographical diversification
- EBITDA split: Germany - 34%, US - 52%, Europe - 14%
- DTE has **unique optionality**, it can pursue different growth modes in different markets. **It is the ultimate defensive stock.**
- Deserves to command a premium for this optionality.

Thesis #3 – Industry head/tailwinds = Consolidation

Macro effects enable large operators [Deutsche Telekom] to deepen competitive positions.

● European Commission Policy Shifts

- The European Commission plans to revise EU competition policy to facilitate mergers and support the green transition
- EU funds for digital transformation, including high-speed internet infrastructure.
- **Telecoms 4 -> 3 operator model was specifically outlined in the Draghi report.**

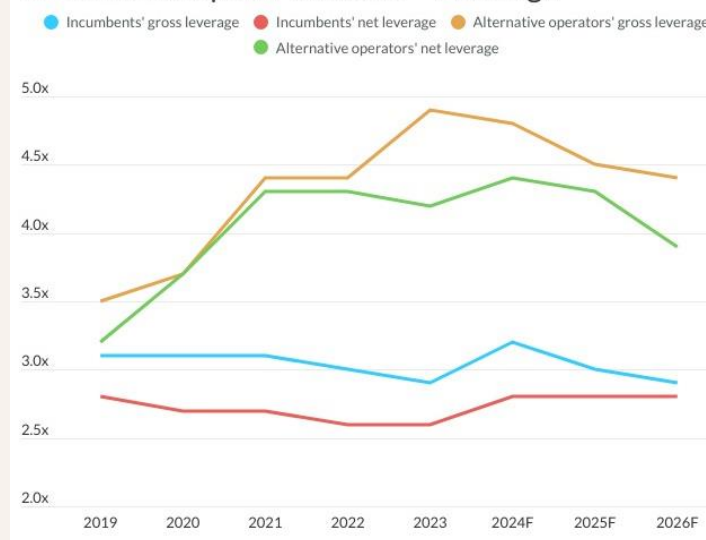
● Post-Pandemic Travel Boom

- Global data roaming market has a projected CAGR of 8.17% (2025 - 2034)
- **DT service revenue growth (4.8% YOY in 2023) is largely driven by increases in roaming**
- Roaming only provided by largest providers.

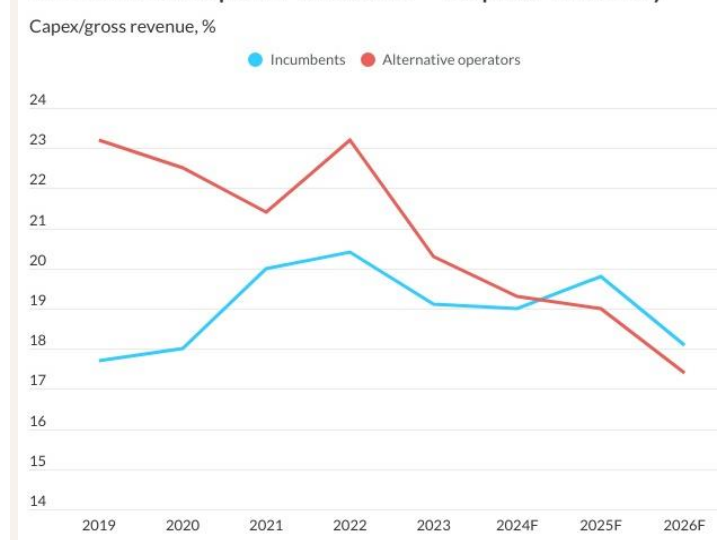
● Interest Rates elevated for longer

- Long-term debt structure allows DT to weather interest costs better than alts.
- Competitors with higher debt burdens will face steeper costs, giving DT competitive advantage. **Highly leveraged alternative net and satellite providers squeezed out.**
- Even among incumbents DT has especially healthy leverage and capex spend.

Western European Telecoms – Leverage



Western European Telecoms – Capital Intensity



Valuation

Undervalued relative to its child company T-mobile, relative to other growth stocks, and intrinsically.

● DCF

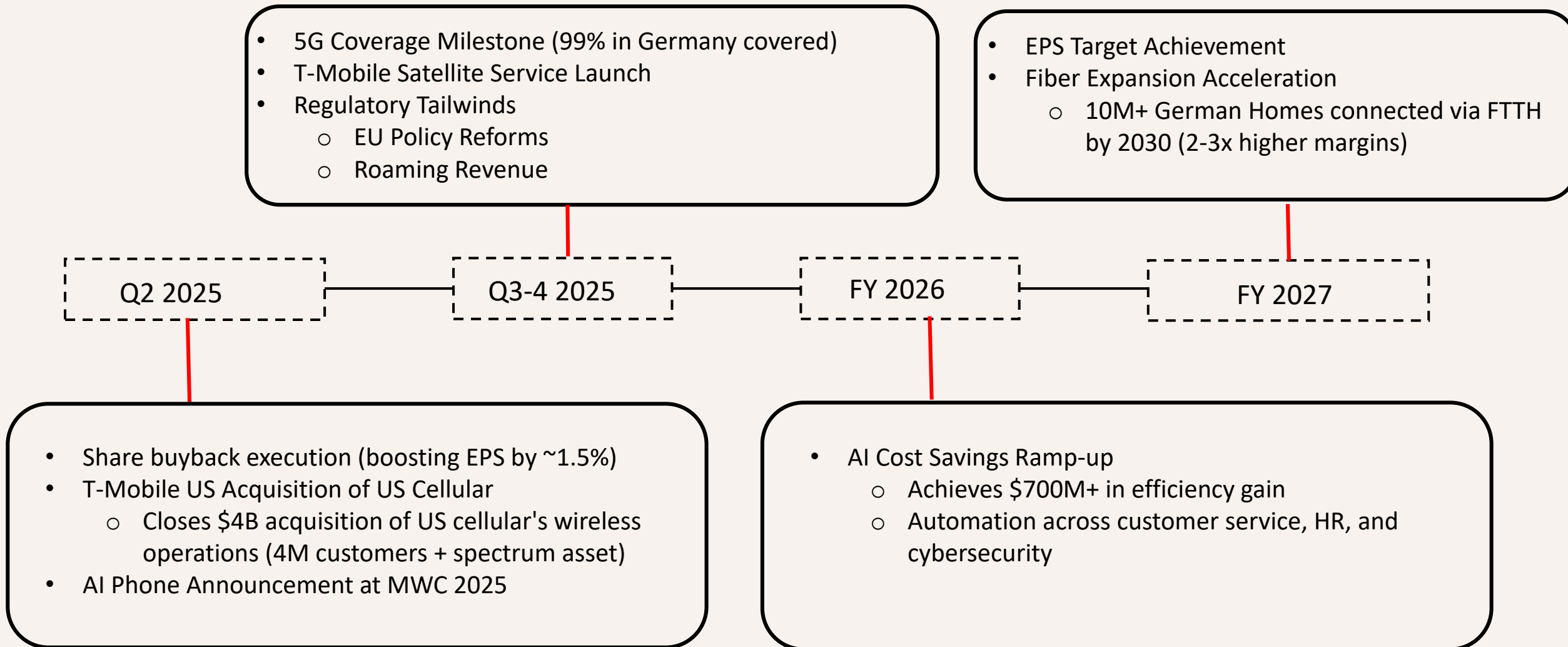
DCF - Base	
Discount Rate	13%
Exit Multiple (EV/EBIT)	10
Present Value	335,773
Net Debt	90,633
Equity Value	245,140
Number of Shares Outstanding	4,892
Implied Share Price	50
Change from Today	38.49%

	Bullish	Base	Bearish
Today's Price	36.18		
Implied Share Price	53.11	50.11	47.60
% Change	46.80%	38.49%	31.57%

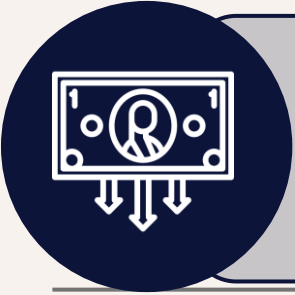
● Public Comps

Company Name	EV/Revenue	EV/EBITDA	EV/EBIT	P/EPS	Forward
Vodafone Group Public	1.8	5.7	17.2	10.2	9.42
Orange S.A.	1.6	4.7	11.0	16.0	10.67
BT Group plc	1.8	5.2	11.7	20.6	8.77
Telefónica, S.A.	1.6	5.5	14.6	NM	13.00
Verizon	2.7	6.7	11.8	11.1	9.85
Telecom Italia S.p.A.	1.3	3.9	12.1	-	NM
Cellnex Telecom, S.A.	10.7	14.2	75.3	NM	NM
Koninklijke KPN N.V.	3.8	8.8	15.1	18.8	15.84
Deutsche Telekom AG	2.9	7.2	12.9	15.2	16.26
High	10.7	14.2	75.3	20.6	15.84
Low	1.1	3.9	11.0	10.2	8.73
Mean	2.8	6.6	19.8	15.4	11.37
Median	1.7	5.6	14.5	15.7	10.26

Catalyst Path

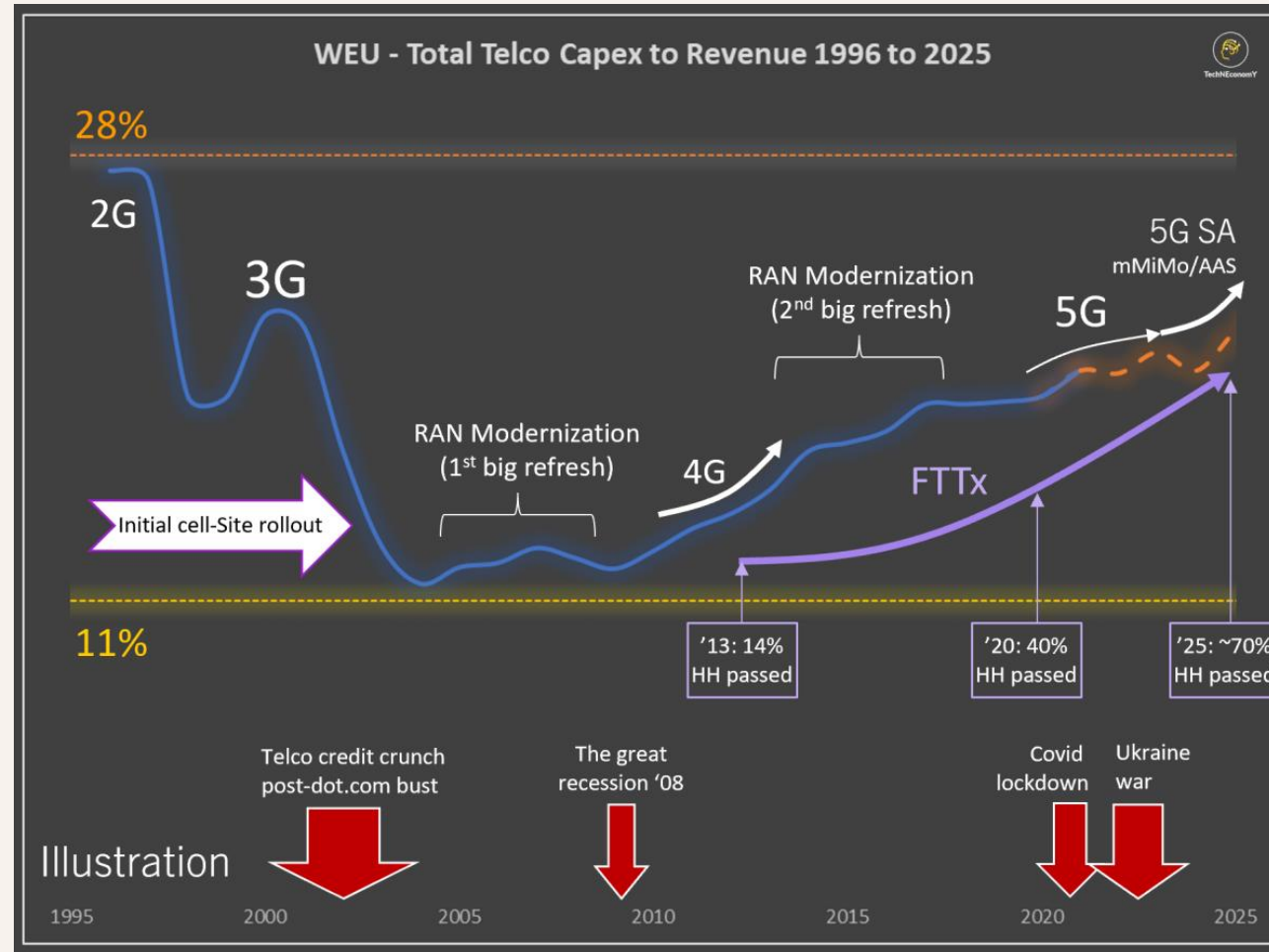


Risks and Mitigants

Risks	Mitigants
 Intense competition from Vodafone, Telefónica, regional providers → market share, pricing impacts <i>E.g. 2021: DT mobile subscription share in Germany ↓ due to Vodafone, MVNO price cuts</i>	Strategic M&A to strengthen position <i>E.g. 2020: Acquired T-Mobile US → expanded global reach, gained North America edge</i>
 Heavy 5G investment (€30B by 2025) → financial strain, risk of delays/underperformance <i>Competition: Huawei, Ericsson → potential market share loss to innovative Chinese competitors.</i>	Continuous R&D, tech partnerships for 5G optimization <i>E.g. Ericsson collab: Industry-leading 5G performance in key EU cities</i>
 Currency fluctuations, economic downturns → impact multinational ops <i>E.g. Q2 2023: 3% profit drop in U.S. due to EUR/USD volatility</i>	Hedging strategies, revenue diversification to offset risk <i>E.g. 2023: New hedge contracts mitigated USD fluctuations</i>

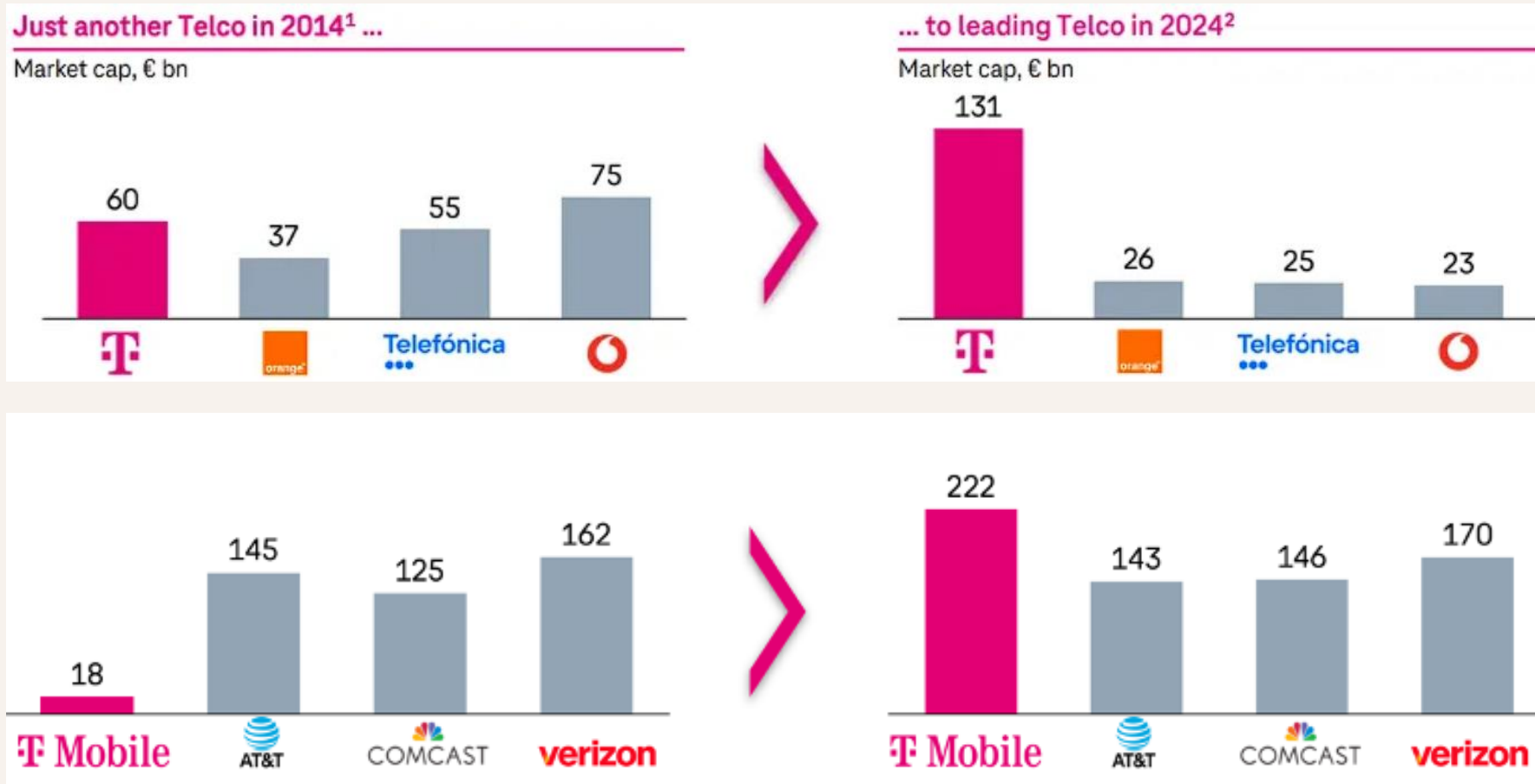
Appendix A

Capex cycle and Secular Demand in the Telco Industry



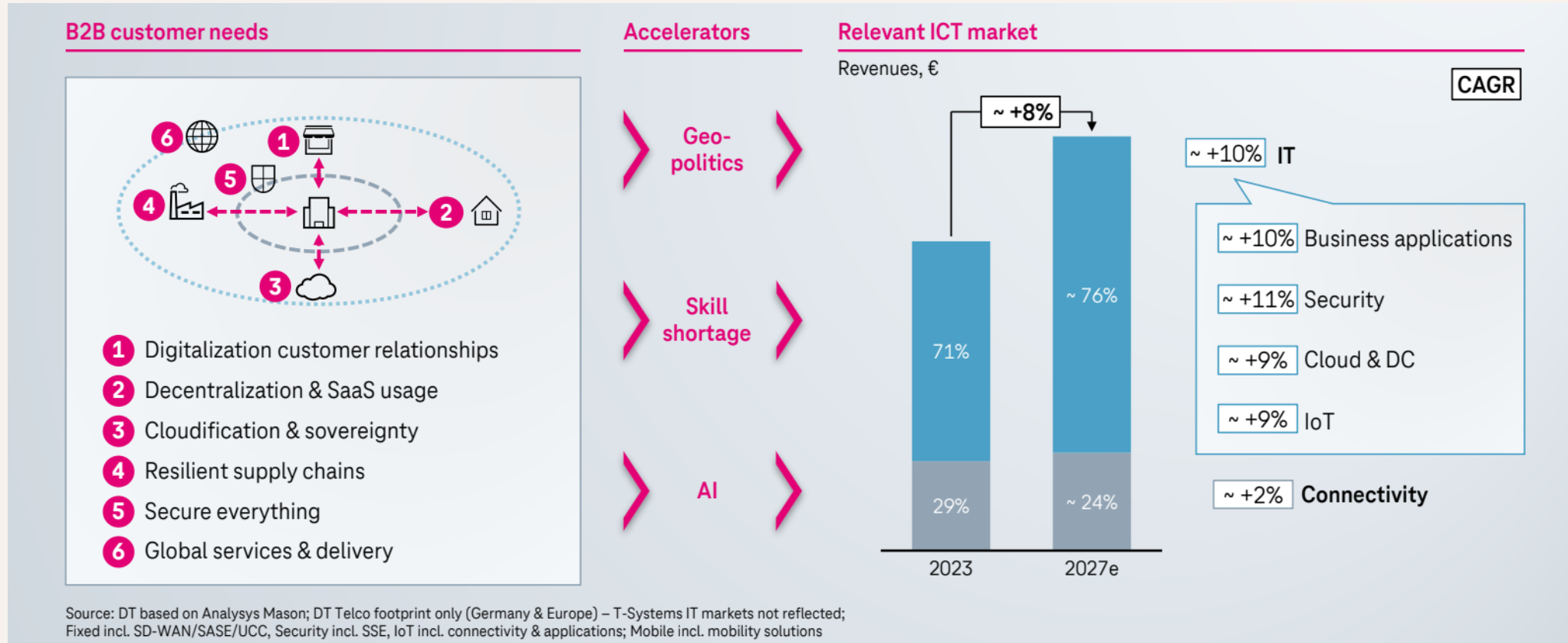
Appendix B

DTE and TMUS have created significant shareholder value



Appendix C

B2B customers set to be growth engine for industry



DTE has over delivered in almost all of its financial targets

CMD 2021 ambition: almost all commitments fulfilled



KPI	Time frame	Ambition as presented (adjusted)	Achievement 2024e ⁴	
Revenues (Group)	2020–2024e	CAGR +1–2%	CAGR ~ 2.1%	✓
Total service revenues (Group)	2020–2024e	CAGR +3–4%	CAGR ~ 3.6%	✓
Adj. EBITDA AL (Group)	2020–2024e	CAGR +3–5%	CAGR ~ 3.3%	✓
Adj. Core EBITDA AL (Group)	2020–2024e	CAGR +5–6%	CAGR ~ 6.9%	✓
Adj. EBITDA AL (DT ex US)	2020–2024e	CAGR +2–3%	CAGR ~ 3.6%	✓
FCF AL (Group)	2024e	> €18 bn (~ €18.3 bn ¹)	~ €19.0 bn	✓
FCF AL (DT ex US)	2024e ³	€4 bn (~ €3.5 bn ¹)	~ €3.5 bn	✓
Adj. EPS (Group)	2024e	> €1.75	> €1.75	✓
ROCE (Group)	2024e	> 6.5%	> 6.5%	✓
Cash Capex (DT ex US)	2024e	~ €8.2 bn (~ €7.6 bn ¹)	~ €7.7 bn	✓
Adj. IDC AL (DT ex US) ²	2020–2024e	Reduction by €1.2 bn	~ €0.0 bn	✗
Leverage (Group) incl. leases	2024e	≤ 2.75x	< 2.75x	✓

T-Mobile beats consensus estimates on growth (T-Mobile US hosted its CMD on 18th September last year).

T-Mobile US

Service revenues:

2023-27E CAGR +5% vs (pre-CMD Bloomberg consensus +4%)

Adj EBITDA:

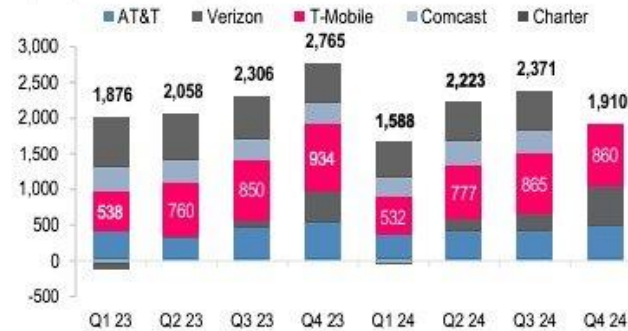
2023-27E CAGR +7% vs (pre-CMD Bbg cons +6.5%)

Appendix F

T-Mobile is the US' fastest-growing telecommunications company and far exceeds competitors.

Figure 1: T-Mobile US continues to take market share

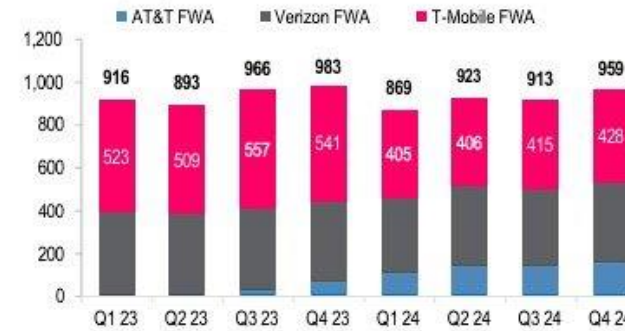
Postpaid phone net adds ('000s)



Source: Company reports. Q4'24 ex Comcast and Charter

Figure 2: T-Mobile US's FWA momentum is even more impressive

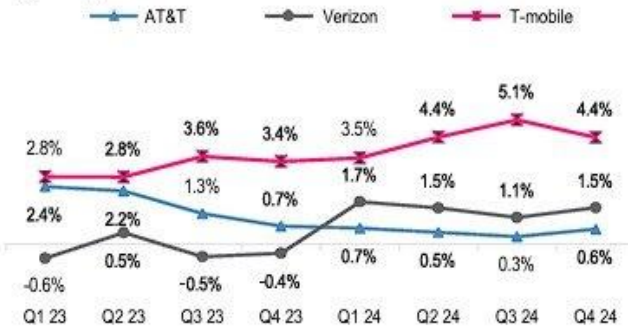
FWA net adds ('000s)



Source: Company reports.

Figure 3: T-Mobile US is delivering MSD service revenue growth

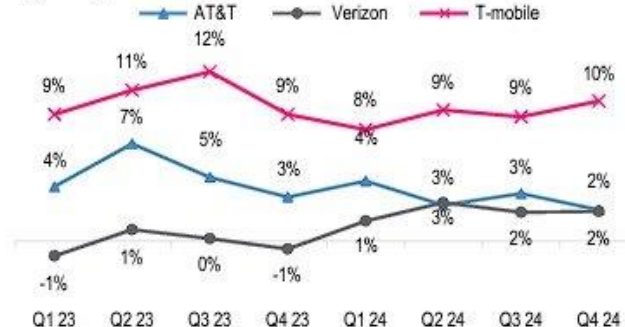
Y/y growth (%)



Source: Company reports.

Figure 4: T-Mobile US's EBITDA performance is even more impressive

Y/y growth (%)



Source: Company reports.

Appendix G

Deutsche Telekom is the most valuable brand in Europe



Appendix H

DCF

		Actual						Forecast				
Income Projection		2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
Revenue	mUSD	93086.07	125624.46	125612.42	125092.44	126772.41	122576.87	128108.79	132495.99	134631.07	135999.60	136749.24
Growth (%)	%		34.96%	-0.01%	-0.41%	1.34%	-3.31%	4.51%	3.42%	1.61%	1.02%	0.55%
Other Revenue	mUSD	28.06	23.24	23.88	26.72	46.42	36.24	30.76	31.21	32.54	33.98	35.19
Growth (%)	%		-17.16%	2.76%	11.87%	73.76%	-21.95%	-15.11%	1.47%	4.26%	4.43%	3.56%
Total Revenue	mUSD	93114.13	125647.71	125636.30	125119.16	126818.83	122613.11	128139.55	132527.20	134663.61	136033.58	136784.43
Growth (%)	%		34.94%	-0.01%	-0.41%	1.36%	-3.32%	4.51%	3.42%	1.61%	1.02%	0.55%
Cost of materials	mUSD	59279.54	75833.64	76415.33	76010.47	71807.23	67791.70	87144.74	92778.40	94274.29	95233.70	95759.66
SG&A	mUSD	-	-	-	-	-	-	-	-	-	-	-
EBITDA	mUSD	27141.73	40287.46	39128.85	36583.31	42250.47	42058.18	40994.81	39748.80	40389.32	40799.88	41024.77
Margin (%)	%	29.16%	32.07%	31.15%	29.25%	33.33%	34.31%	32.00%	30.00%	30.00%	30.00%	30.00%
D&A	mUSD	19380.54	30072.17	30482.20	28298.60	26256.22	24749.97	20503.10	22626.53	21064.82	21845.68	21455.25
Other Operating Exp/Income	mUSD	2597.91	3694.19	3342.43	5264.51	4444.57	4811.06	3075.46	3393.98	3159.72	3276.85	3218.29
EBIT	mUSD	11856.13	16047.71	15396.34	15545.58	24310.82	25260.38	29327.00	28817.91	26012.26	24916.68	22970.37
EBIT Margin	%	12.74%	12.77%	12.26%	12.43%	19.18%	20.61%	22.89%	21.75%	19.32%	18.32%	16.80%
Tax	mUSD	2236.56	2272.78	1969.75	2070.11	4058.80	5488.15	7331.75	7204.48	6503.07	6229.17	5742.59
Effective Tax Rate	%	27.45%	21.97%	22.71%	17.65%	30.68%	23.09%	25.00%	25.00%	25.00%	25.00%	25.00%
EBIAT(NOPAT)	mUSD	8601.86	12522.46	11900.48	12802.17	16851.82	19427.78	21995.25	21613.43	19509.20	18687.51	17227.78
(+) D&A	mUSD	19380.54	30072.17	30482.20	28298.60	26256.22	24749.97	20503.10	22626.53	21064.82	21845.68	21455.25
Capital Expenditure	mUSD	(11201.90)	(15826.30)	(15485.00)	(17701.20)	(13602.30)	(11593.30)	(9298.93)	(11498.18)	(10796.80)	(10531.30)	(10942.10)
Change in NWC	mUSD	2510.38	3876.45	5324.69	(2306.30)	872.11	766.12	1819.12	1292.62	1555.87	1424.25	1490.06
Unlevered FCF	mUSD	8168.70	13722.78	4798.42	14553.01	21680.39	20697.15	26380.30	28949.17	28221.34	28577.64	26250.87
Discounting Factor	%	1	1	1	1	1	1	0.87	0.76	0.66	0.57	0.50
								22950.86	21911.63	18583.84	16372.06	229703.69
												255954.56